

Structured Job Task Update in Maximo Using Job Plan Number

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1. Background

In IBM Maximo, a **job plan** is identified by a unique **job plan number**, which serves as a reference for predefined sets of maintenance tasks. Each job plan consists of multiple **job tasks**, where each task represents a specific step or action, typically stored in the description field.

In many cases, job plans are created with a **single long description** or a **high-level instruction** under the plan, rather than a structured breakdown of individual steps. This makes it harder for technicians to follow the procedure in a clear, step-by-step manner during execution.

This use case aims to improve the clarity and usability of existing job plans by:

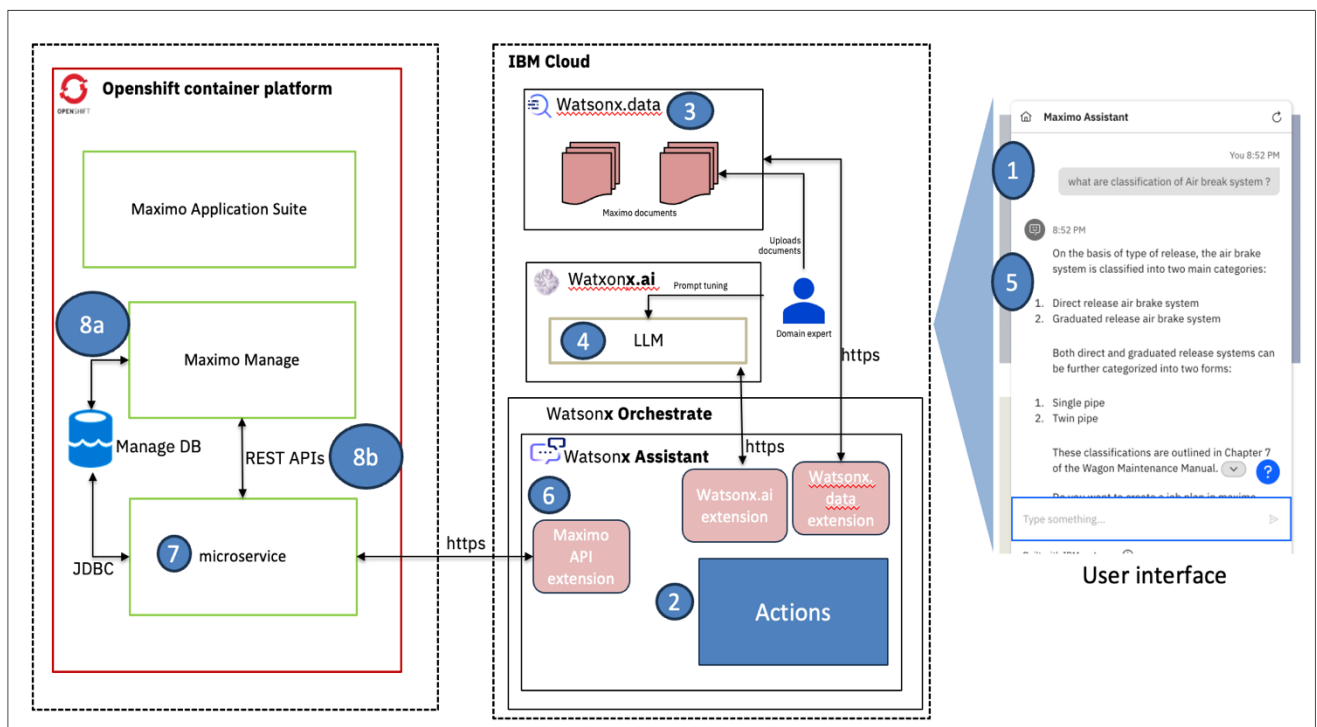
- Retrieving the job plan using the **provided job plan number**
- Analysing the long, unstructured description
- **Breaking it down into smaller, individual job tasks**
- Updating the Maximo job plan with this structured list of tasks

By transforming vague or condensed instructions into actionable task items, this approach:

- Enhances technician understanding and execution
- Improves compliance with procedures
- Reduces the chances of human error
- Aligns job plans more closely with real-world operational workflows

This solution leverages automation and integration with Maximo to streamline the update process—ensuring job plans are not only technically accurate but also user-friendly and execution-ready.

2. Architecture Diagram



3. Technologies used:

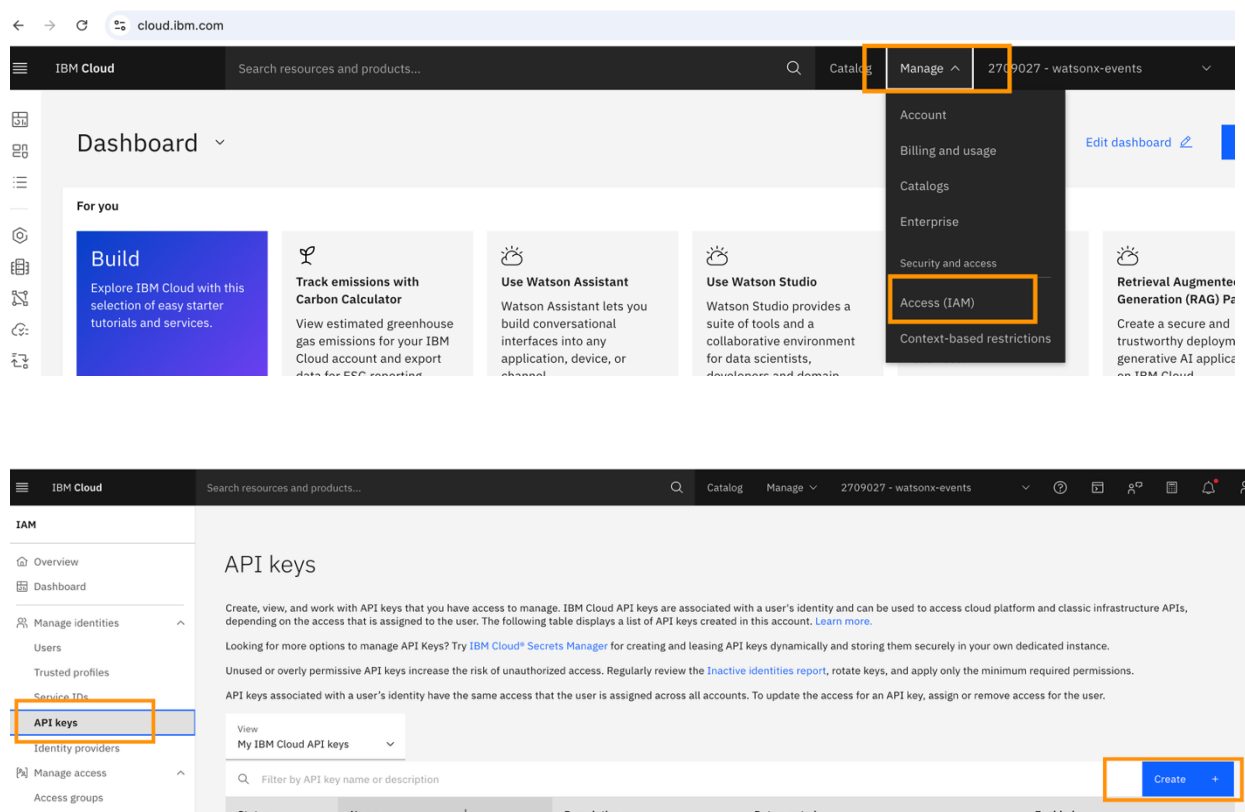
Services	Purpose
Watsonx.ai Studio	To generate structured responses using LLMs
Watsonx.ai Runtime	To host and run foundation models
Watsonx Assistant	For building conversational search applications
Maximo	Asset Management Tool

4. Create API key

Why to Create API key: An API key is required to securely authenticate and authorize communication between services in your Watsonx and Maximo environment.

- Always keep API keys confidential.
- Rotate them periodically.
- Never hard-code them directly into source code (use .env or secrets managers instead).

4.1 Go to **IBM Cloud** → click on **Manage** → click on **Access (IAM)** → then click on **API Keys** on the left side → Click on **Create**



4.2 Provide **Name** and **Description** → Click on **Create** → Copy & download the API key for future reference

Create IBM Cloud API key

Name

api-key

Description (optional)

Enter description

Leaked action

If API key is discovered to have been leaked out in the world, what would you like the system to do?

☒ Disable the leaked key
 ☐ Delete the leaked key
 ☐ Nothing

Session creation

Will this API key create a session in the CLI?

☐ Yes
 ☒ No

Cancel

Create

5. Watsonx.AI Configurations

5.1 Launch watsonx.ai

Click on *Resource list* from left side panel in *IBM Cloud* → Click on *watsonx.ai Studio* under *AI / Machine Learning* resource. (highlighted in image)

Name	Group	Location	Product	Status	Tags
AI / Machine Learning (11)					
Watson Orchestrate-itiz	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx Orchestrate	Active	
Watson Orchestrate-itiz	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx Orchestrate	Active	
wa-itiz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx Assistant	Active	
watsonx.ai Runtime-pe	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Runtime	Active	cpdaas
wml-itiz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx.ai Runtime	Active	
wml-itiz-wxo-663003CR67_67efa2b5dbb	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Runtime	Active	
ws-itiz-wxd-68222c2ee52786c3f70bdc	itz-wxd-68222c2ee52786c3f70bdc	Dallas (us-south)	watsonx.ai Studio	Active	
ws-itiz-wxo-663003CR67_67efa2b5dbb	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx.ai Studio	Active	

5.2 Select Launch in *IBM watsonx*

IBM Cloud

Search resources and products...

Catalog

Resource list /

ws-itiz-wxo-663003CR67_67efa2b5dbb

Manage

Plan

watsonx.ai Studio in

Cloud Pak for Data and watsonx

Build and deploy machine learning models on either platform.

Work with foundation models on watsonx as a Service.

Launch in

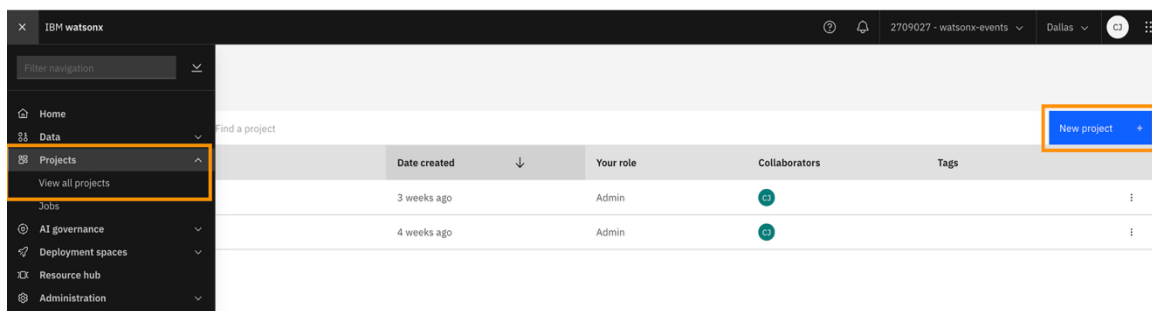
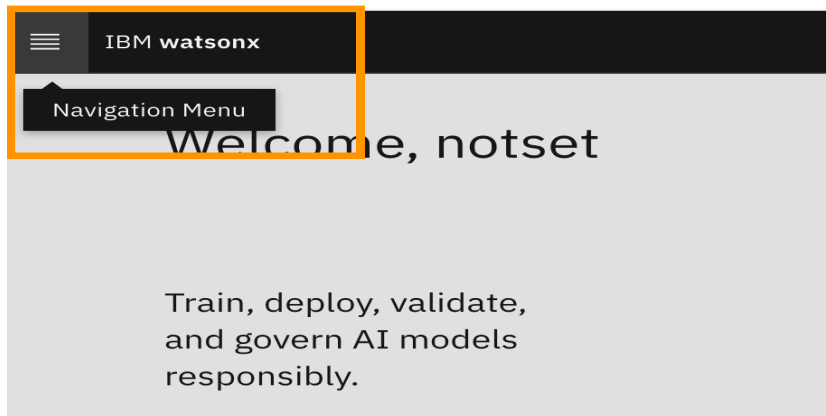
IBM Cloud Pak for Data

IBM watsonx

Helpful links

5.3 Create New project

Click on **Navigation Menu** at top left side → Click on **Projects** → Click on **View all projects** → Click on **New project**.



5.4 Provide **Name** and **Select storage service** from drop down → Click on **Create**.

Create a project

Start with a new, blank project or select from where to import an existing project.

+ New

Local file

Sample

project1

Enter a name

Description (optional)

What's the purpose of this project?

Tags (optional)

Add tags

Add tags to make projects easier to find. To add tags, separate them with commas and press Enter.

Define storage

✓ Select storage service

Target Cloud Object Storage Instance

cos-itz-wxo-663003CR67_67efa2b5dbb

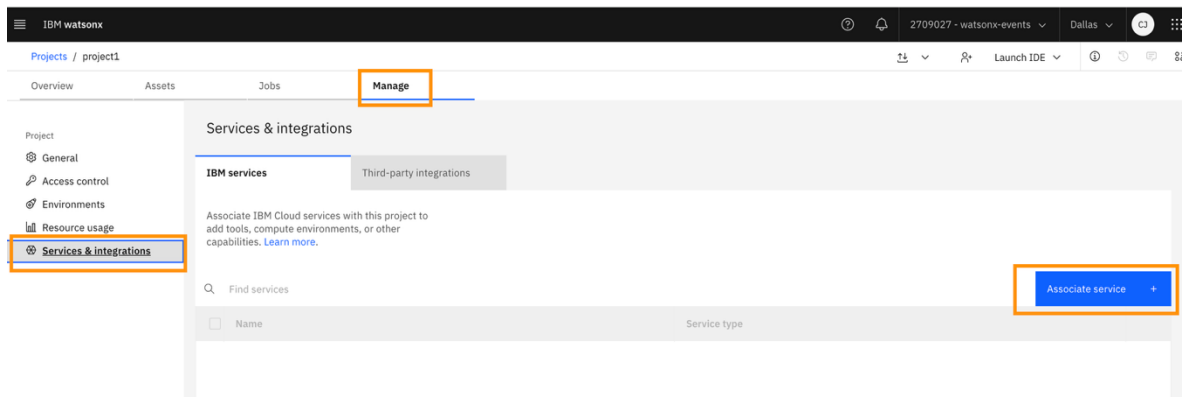
Project includes integration with [Cloud Object Storage](#) for storing project assets.

Advanced settings

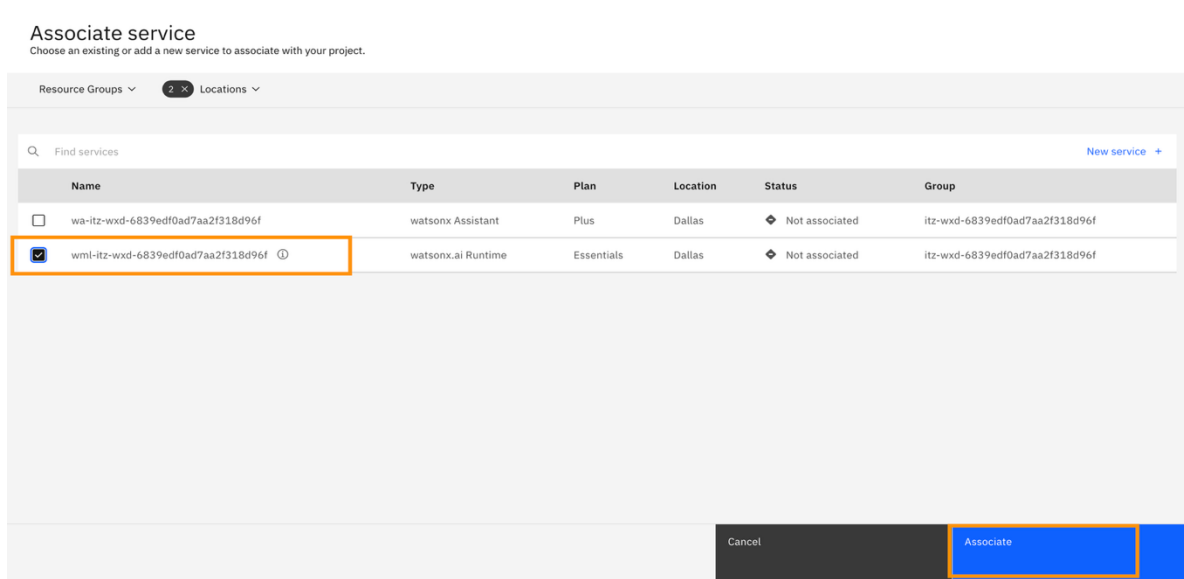
Cancel

Create

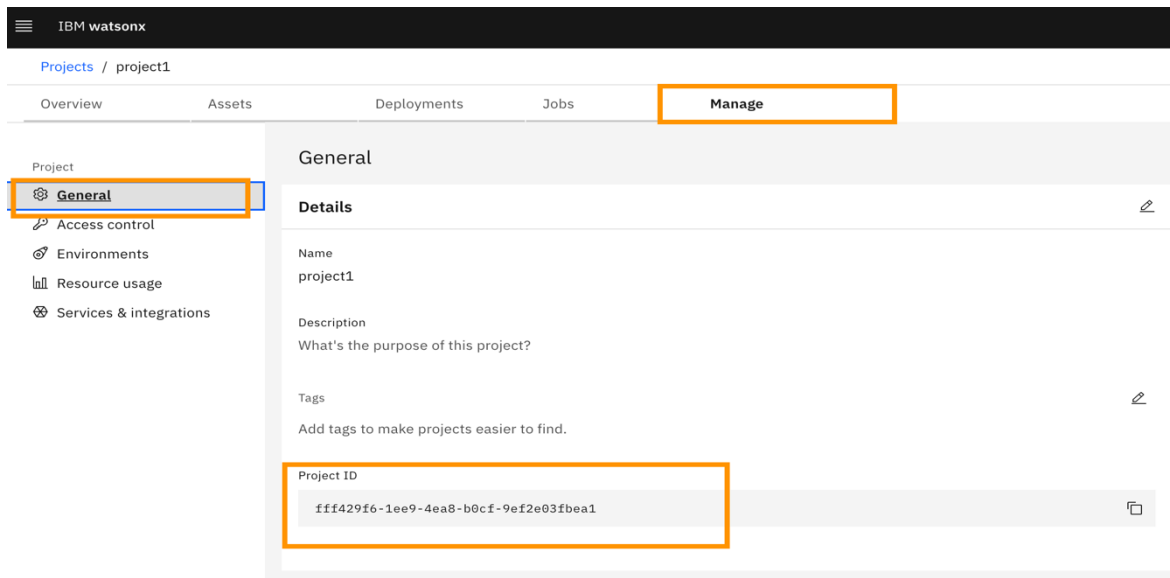
5.5 Once project created, Navigate on **Manage** tab → select **Services & Integrations** (left side panel) → click on **Associate service** (highlighted in image)



5.6 Select service name starts with wml(ex: wml-xxx-xxx-xxxxxxx) → click on *Associate*.



5.7 Navigate to *Manage* tab → Click on *General* → Copy and Save *Project ID* for the future use



6. Watsonx Assistant Configurations

6.1 Launch watsonx assistance in orchestrate

Go to Resource list in *IBM Cloud* → Expand *AI/Machine learning* → Click on *Watson Orchestrate-*

(highlighted in image)

Resource list

Name	Group	Location	Product	Status	Tags
Filter by name or IP address... Filter by group... Filter... Filter... Filter... Filter...					
Compute (0)					
Containers (4+)					
Networking (0)					
Storage (2+)					
Converged infrastructure (0)					
Enterprise applications (0)					
AI / Machine Learning (11)					
Watson Knowledge Catalog-itz	watsonx	London (eu-gb)	watsonx.data intelligence	Active	donotde... +1
Watson Knowledge Catalog-itz	watsonx	Dallas (us-south)	watsonx.data intelligence	Active	donotde... +1
Watson OpenScale-itz	watsonx	Dallas (us-south)	watsonx.governance	Active	donotde...
Watsonx Orchestrate-itz	itz-wxd-00222c2e92e700c3770bd...	Dallas (us-south)	watsonx Orchestrate	Active	—
Watsonx Orchestrate-itz	itz-wxo-663003CR67_67efa2b5dbb	Dallas (us-south)	watsonx Orchestrate	Active	—

6.2 Click on *Launch watsonx Orchestrate*

Resource list /

Watson Orchestrate-itz

Manage

- Service credentials
- Plan

Start by launching the tool

[Launch watsonx Orchestrate](#) [Getting started tutorial](#)

Credentials

API key: [Download](#) [Show credentials](#)

URL: <https://api.us-south.assistant-builder.watson.cloud.ibm.com/instances/706c...>

Plan

Esse

Upgrade

6.3 Click on *Assistant builder* within *Build* section on the Left Hamburger menu

IBM watsonx Orchestrate

English Contact Sales Buy Now

New chat

10:49 AM

Hello, welcome to watsonx Orchestrate

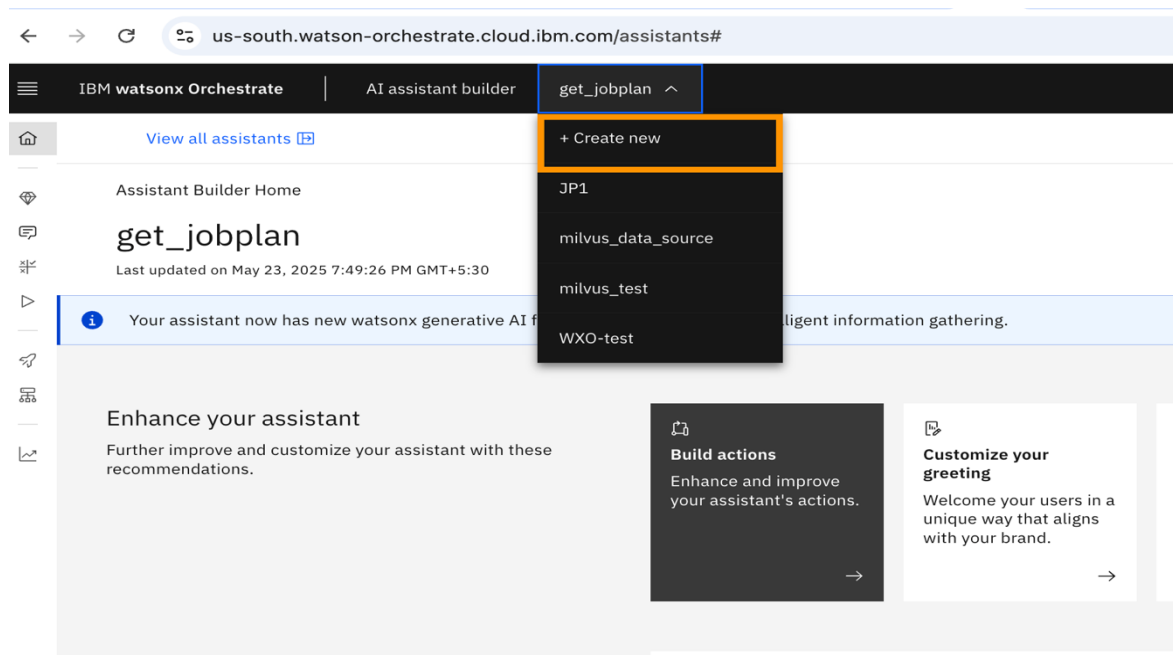
Accuracy of generated answers may vary. Please double-check responses.

What can you do for me?
Learn about watsonx Orchestrate AI Agent and Gen AI capabilities

Formalize Message
Helps you craft more professional messages

Summarize Meeting Notes
Bullet key points from meeting notes

6.4 Click on drop down next to the *AI assistant builder* → Click on *Create new* for creating new Assistant



Provide *Assistant details* (ex: get_jobplan)

The screenshot shows the 'Create your first assistant' form in the IBM watsonx Orchestrate AI assistant builder. The form is titled 'Create your first assistant' and includes fields for 'Assistant name' (pre-filled with 'get_jobplan'), 'Description (optional)' (with a placeholder 'Add a description for this assistant'), and 'Assistant language' (pre-filled with 'English (US)'). The form is part of a multi-step process with tabs for 'Create', 'Personalize', 'Customize', and 'Preview'.

Welcome to AI assistant builder

Back

Next

Create

Personalize

Customize

Preview



Personalize your assistant

Tell us where your assistant will live

You may add multiple channels from your dashboard.

Where do you plan on deploying your assistant?

I'm not sure

No problem! Let's get you started with web chat.

Tell us about yourself

This information will be used to personalize your onboarding experience.

Which industry do you work in?

Software

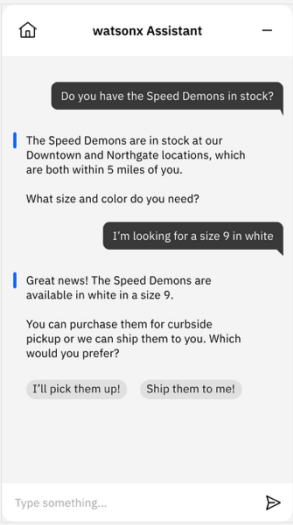
What is your role on the team building the assistant?

Developer

Which statement describes your needs best?

I want to provide confident answers to common questions

This is what your customers will experience



Welcome to AI assistant builder

Back

Next

Create

Personalize

Customize

Preview



Customize your chat UI

Update the style to match your brand and your website. You can change these settings later. A developer can also add more advanced styling changes with code. [Learn more](#)

Assistant's name as known by customers

Assistant

Intended purpose

- ☒ Standard: For virtual agents and customer support experiences.
- ☐ Carbon for AI: For use in internal IBM products.

Choose a theme

Light

Dark

Primary color

#FFFFFF

Chat header

Accent color

#0354E9

Significant and interactive objects

Size

The size of the web chat on this page will not change by updating these fields.



Add an avatar image

Restart conversation

Assistant

Hi! I'm a virtual assistant.
How can I help you today?

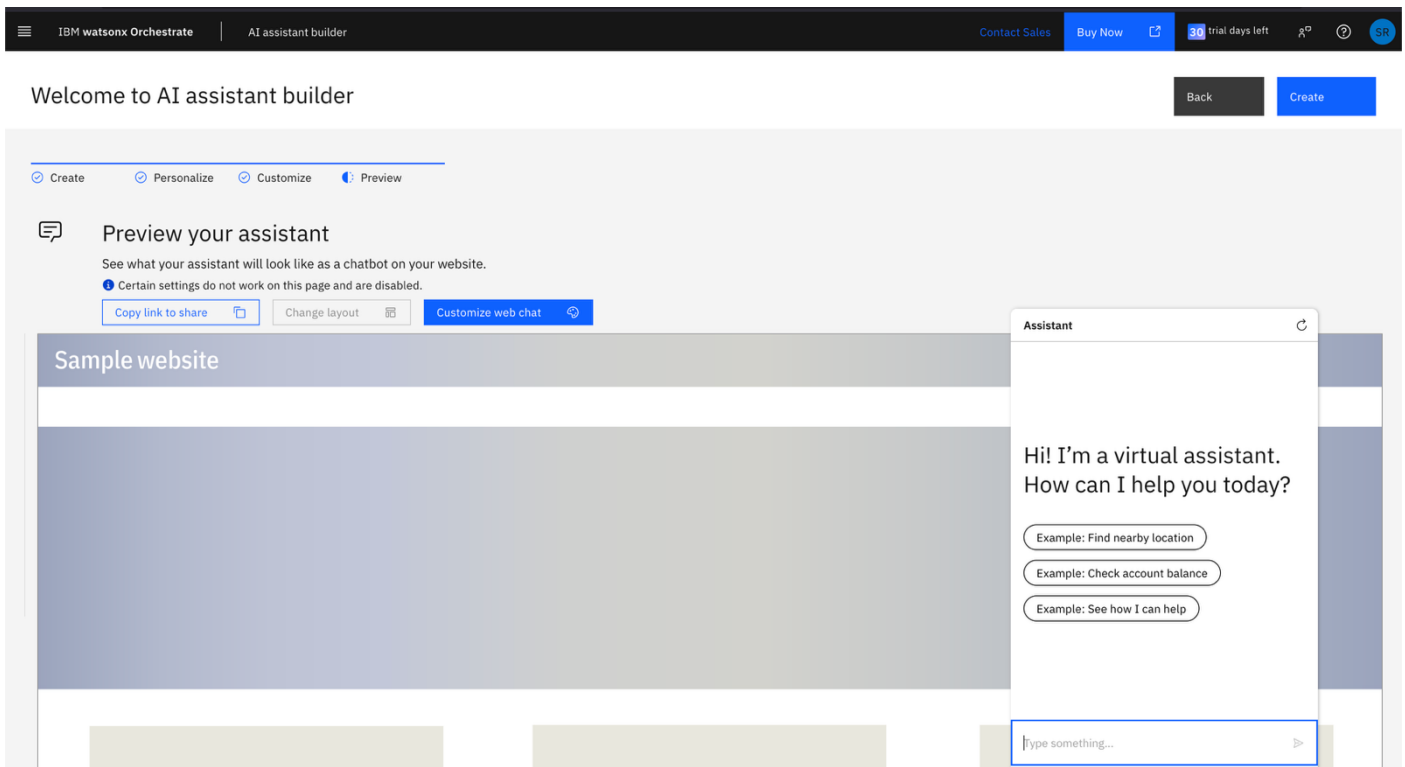
Example: Find nearby location

Example: Check account balance

Example: See how I can help

Type something...

Built with IBM watsonx

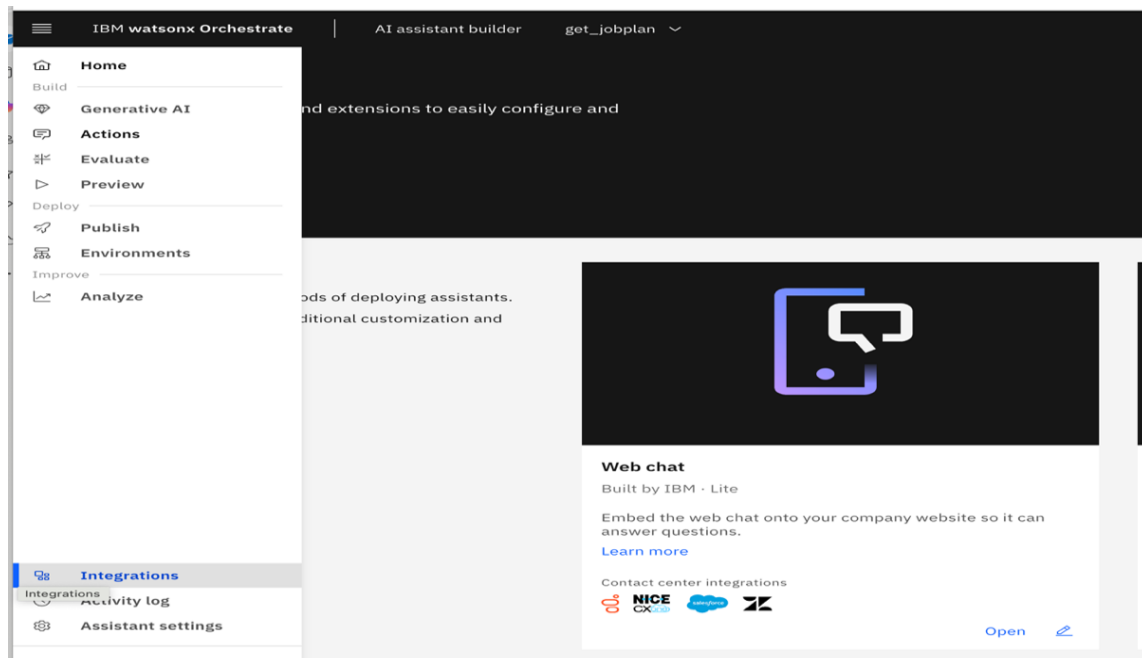


7. Adding Custom Extensions

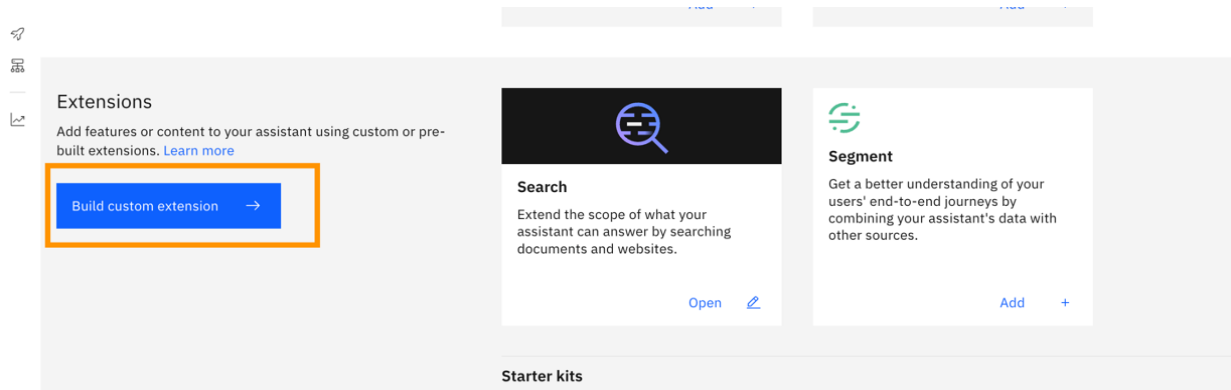
7.1 Watsonx AI integration

7.1.1 Create Api key in *IBM Cloud* for authentication → follow Step3 to *Create API key*.

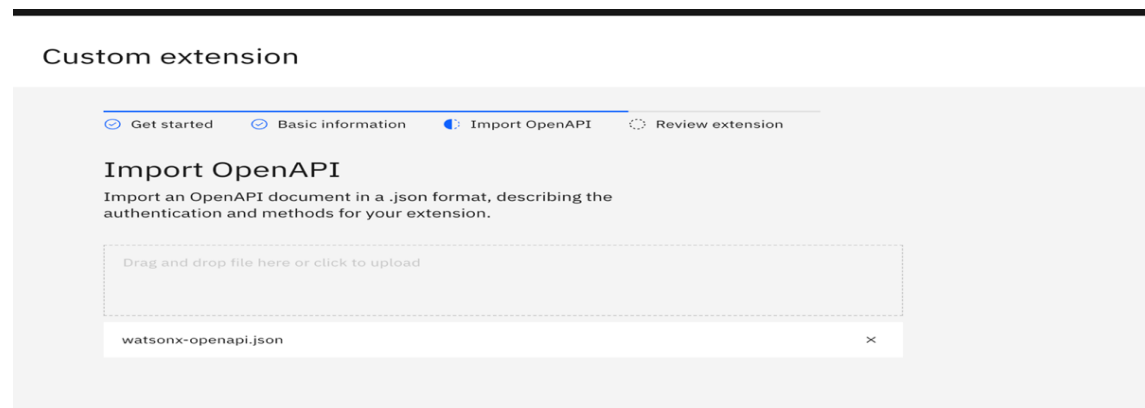
7.1.2 From Left Navigation, select *Integrations* option



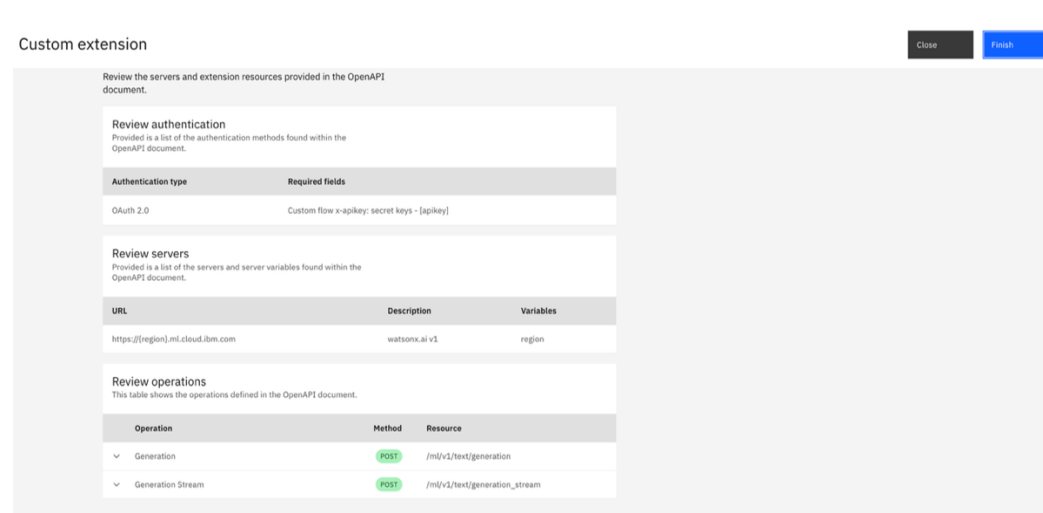
7.1.3 Click on *Build custom extension*



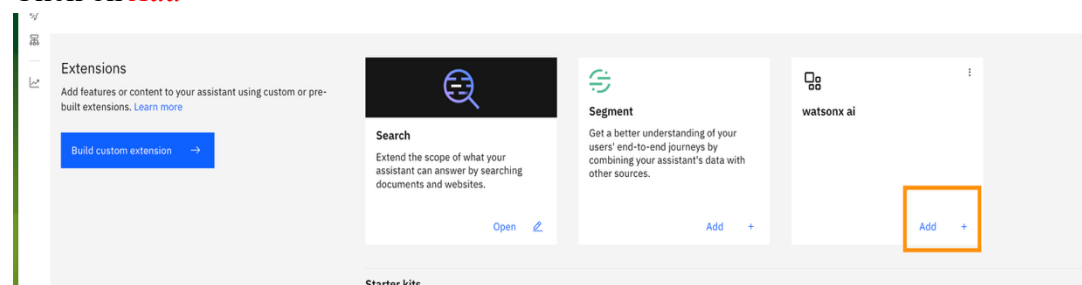
7.1.4 Use watsonx-openapi.json OpenAPI specification & import it.



7.1.5 Review and click on *Finish*



7.1.6 Click on *Add*



Custom extension Draft Close Next

Get started **Authentication** Review operations

Authentication

Authentication types are determined in the OpenAPI document and provide security for the extension.

Authentication type: OAuth 2.0

Grant type: Custom apikey

Custom Secrets

Apikey: *****

Client authentication ⓘ

Send as Body

Header prefix ⓘ

Bearer

Servers

https://{region}.ml.cloud.ibm.com

Generated URL: https://us-south.ml.cloud.ibm.com

Server variables

Enter values for your server variables to create a valid generated URL for requests.

region: us-south

7.1.7 Select/Provide below fields

- Authentication type: OAuth 2.0
- Grant type: Custom apikey
- Api key: provide apikey created in step 6.1.1
- Client authentication: sent as body

Custom extension Draft Close Finish

Get started Authentication **Review operations**

Review operations

This table shows the operations defined in the OpenAPI document.

Operation	Method	Resource
Generation	POST	/ml/v1/text/generation
Generation Stream	POST	/ml/v1/text/generation_stream

7.1.8 Click on next & *Finish*

7.2 Maximo custom extension

We need to build and deploy code in Maximo to support the following three scenarios

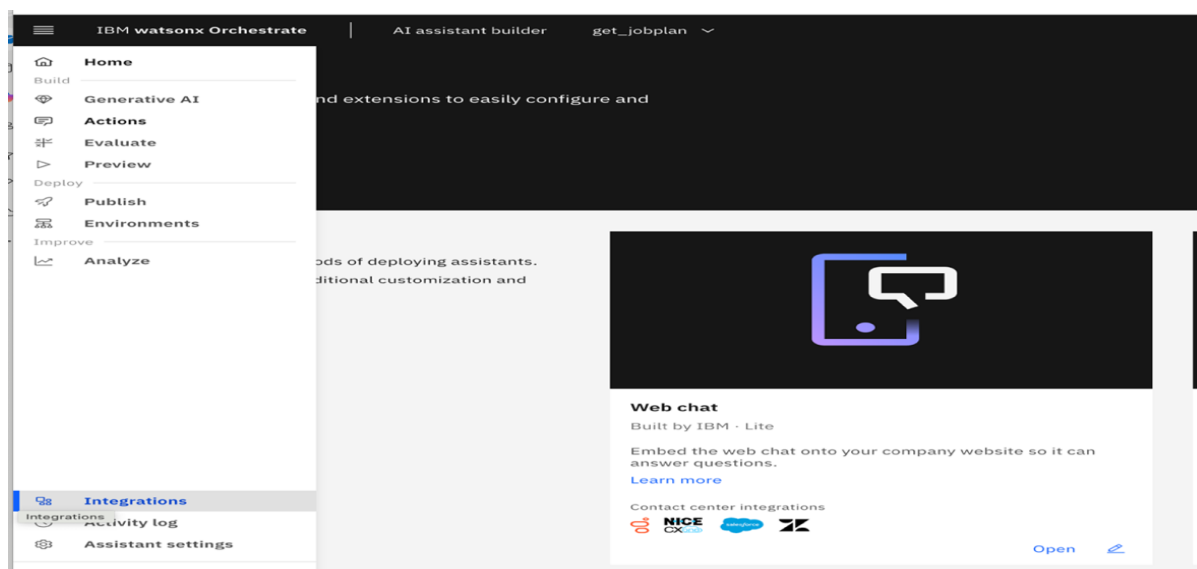
- Retrieve job plan details
- Update job plan status
- Update job plan tasks

The steps outlined below apply to *one(Retrieve job plan details)* scenario. Please follow the same steps for the remaining two scenarios as well.

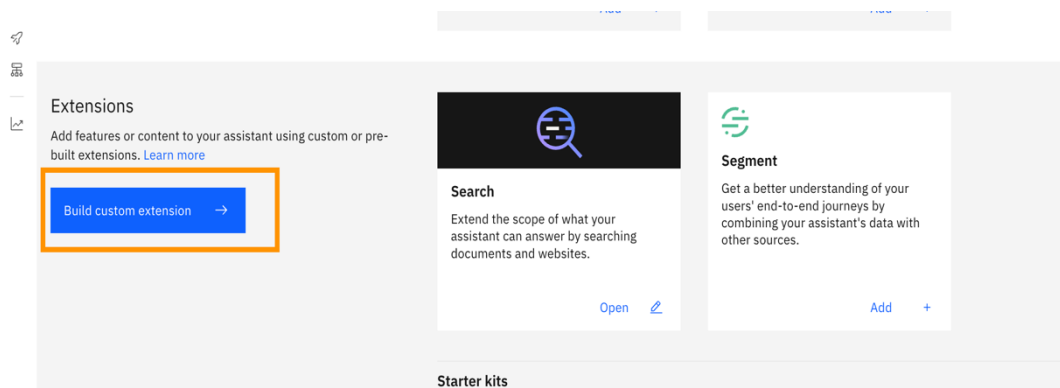
7.2.1 Download the optimize_jobplan.json OpenAPI file.

7.2.2 Use **optimize_jobplan.json** OpenAPI file to add Maximo extension to retrieve the job plan details for the given job plan number.

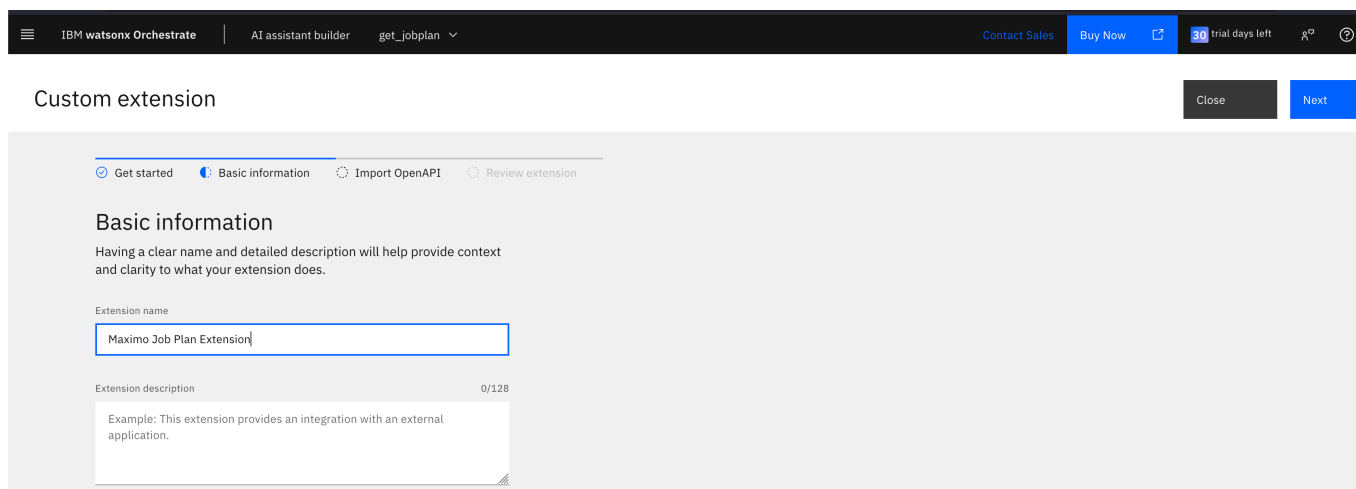
From Left Navigation, select *Integrations* option

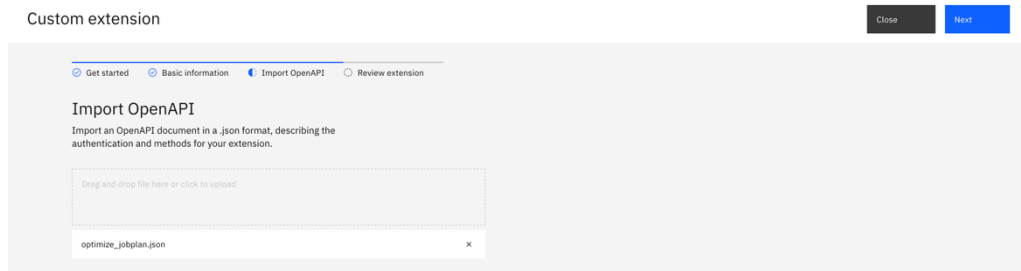


7.2.3 Click on *Build custom extension*

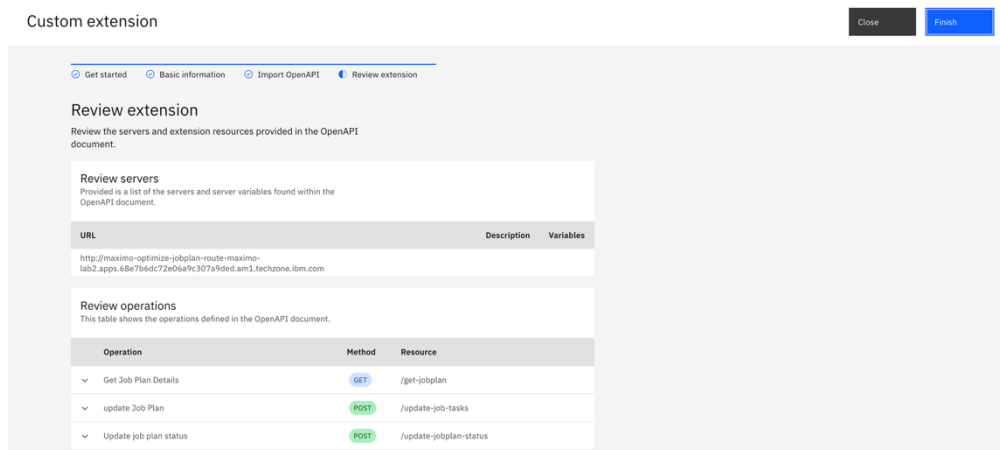


7.2.4 *Import open API* file

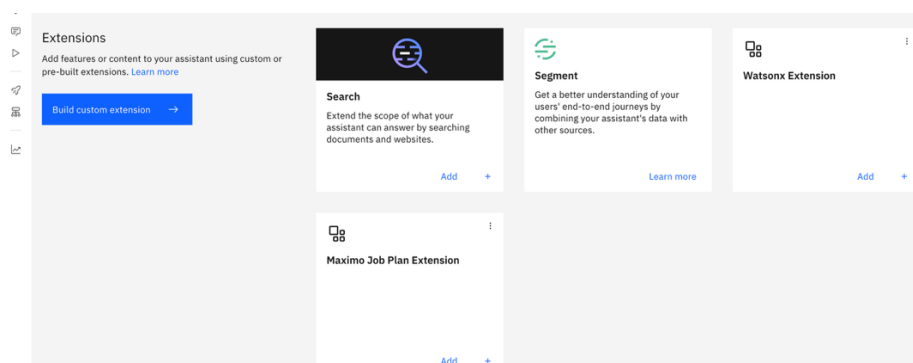




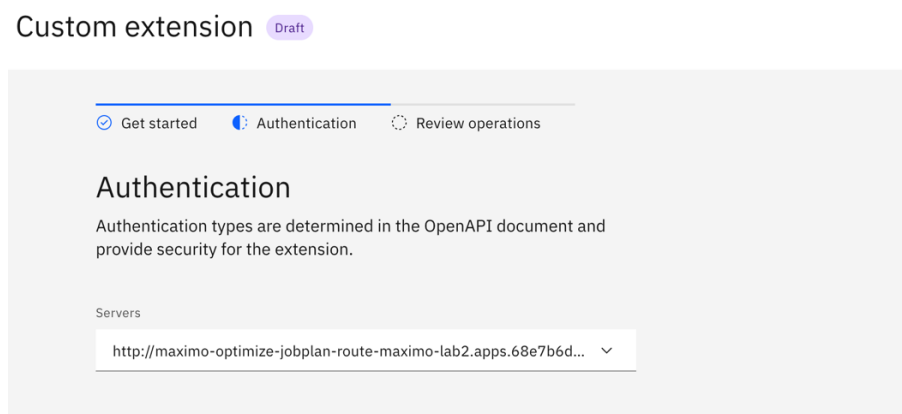
7.2.5 Review and click on *finish*



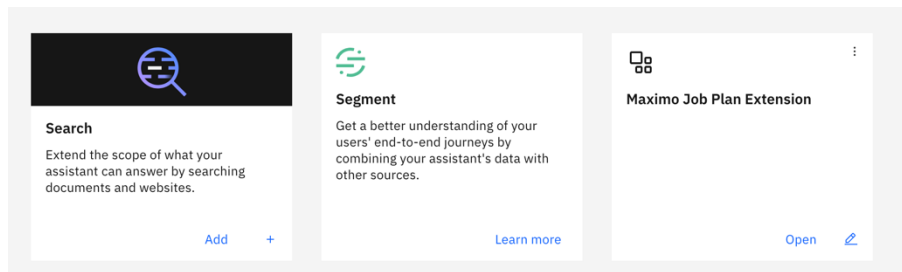
7.2.6 Click on *Add*



7.2.7 Make sure under *Authentication* you can see the Maximo URL (same URL will be present in the openAPI file)



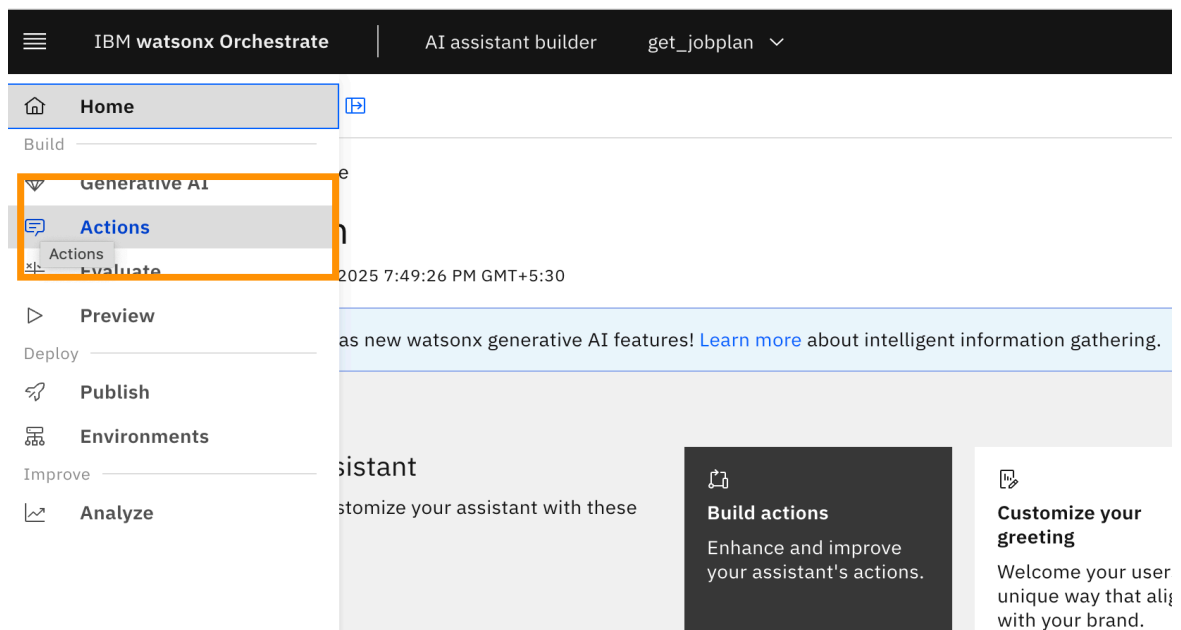
7.2.8 Click Next and *Finish*.



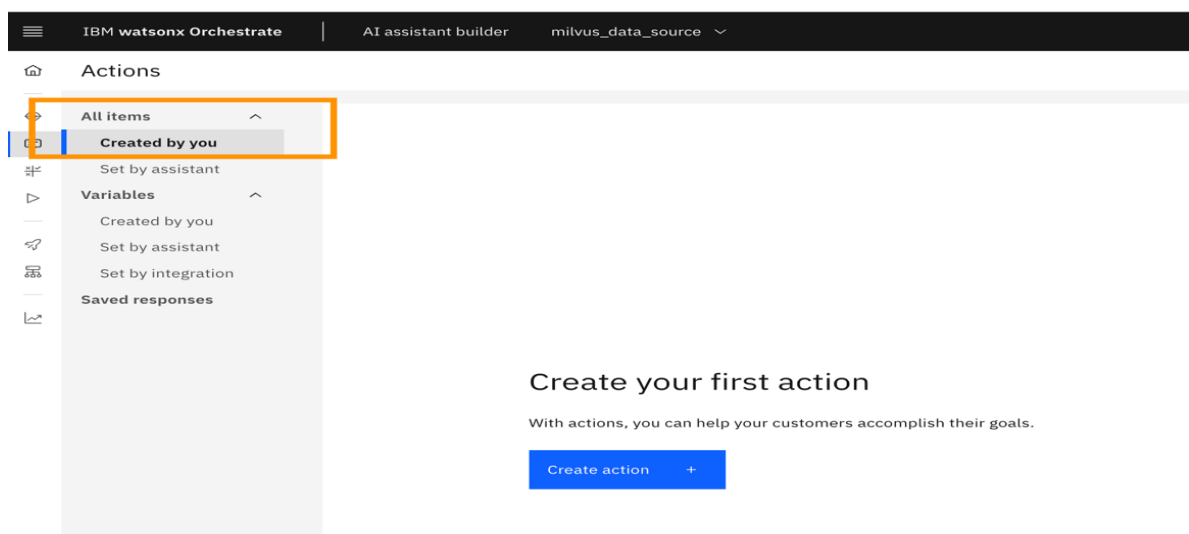
8. Import Actions

8.1 Download the file [actions.json](#)

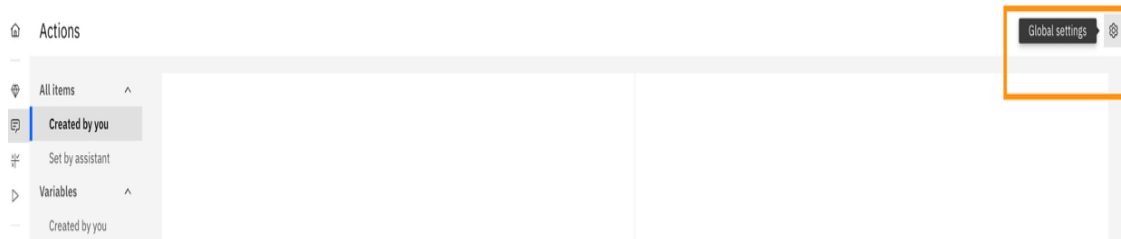
8.1.1 Select *Actions* form the left side panel



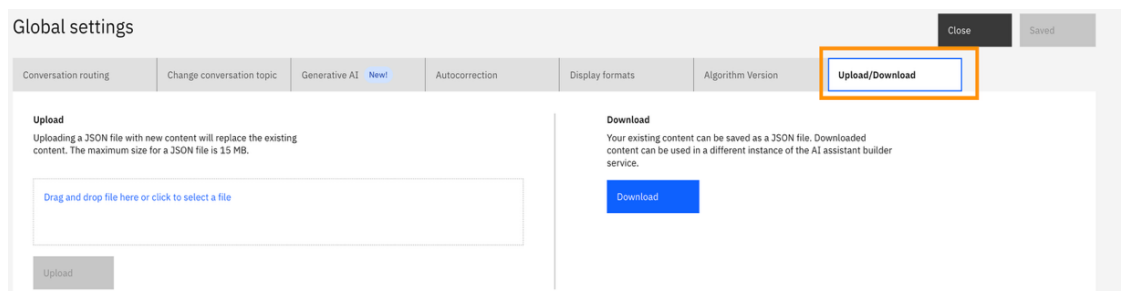
8.1.2 Click on *Created by You*



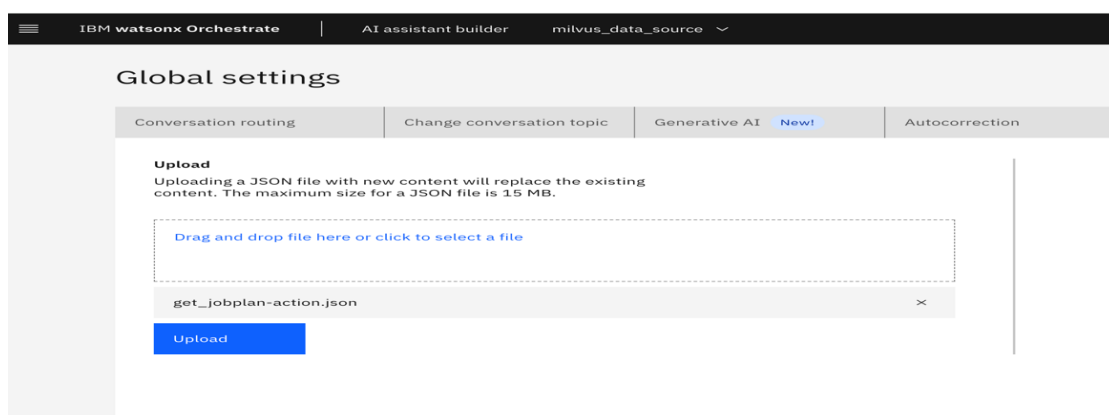
8.1.3 Select *Global settings* on right top corner



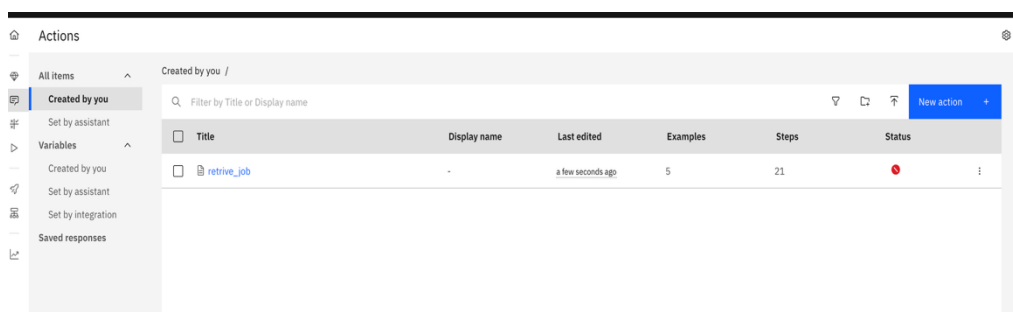
8.1.4 Navigate to *Upload/Download* option



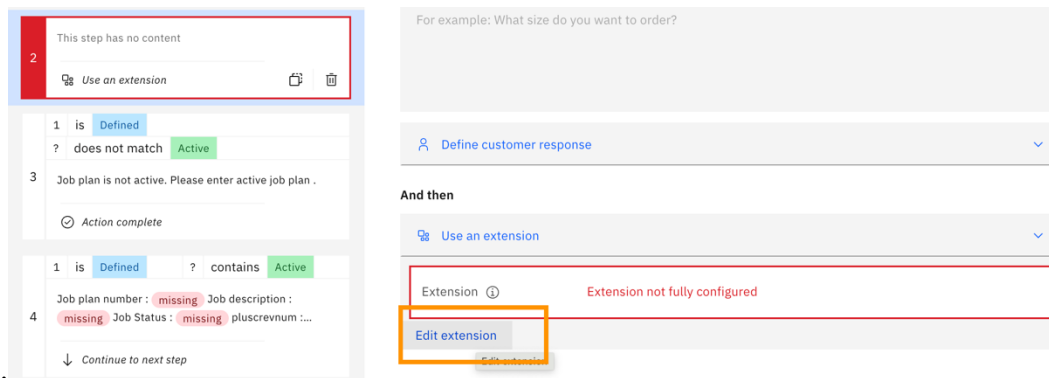
8.1.5 Click on *Drag and drop file here* option & select *actions.json* file → Click on *Upload*



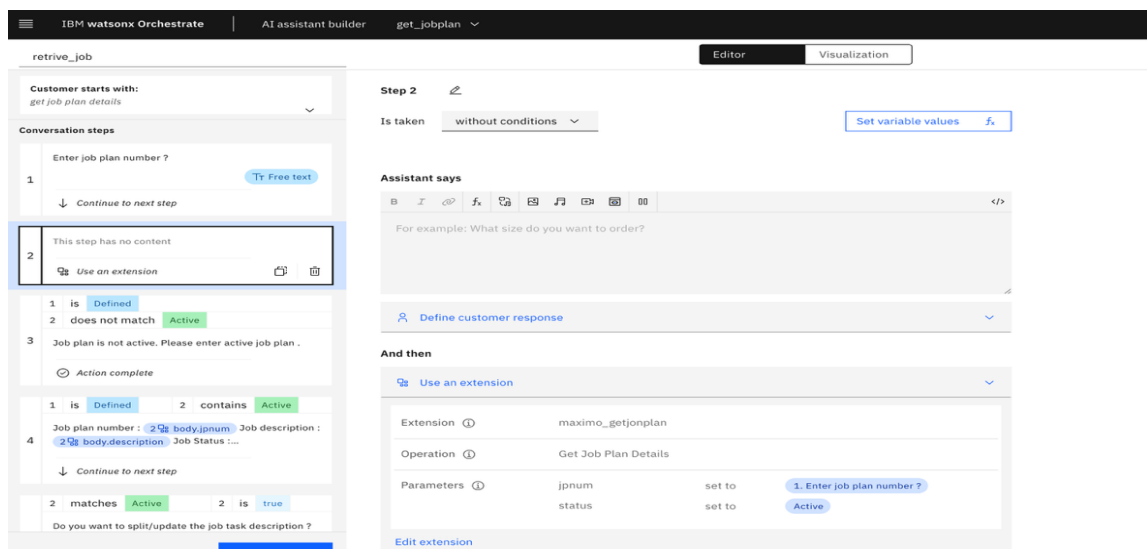
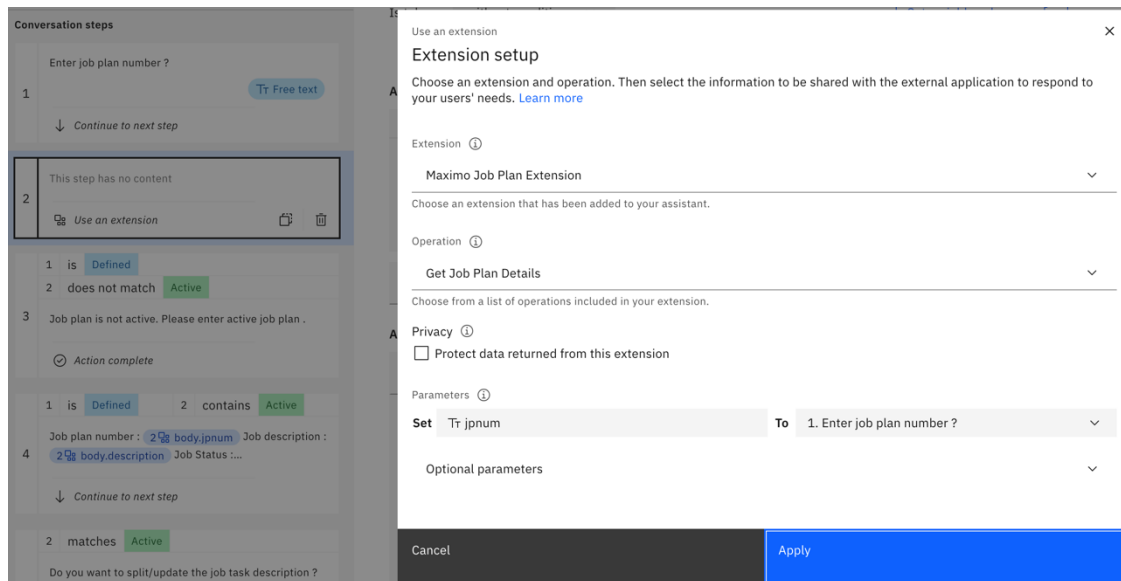
8.1.6 Click on *retrive_job* title



8.1.7 Click on Step2 and select *Edit settings* → select **Maximo Job Plan Extension** Maximo extension



Set *Status to Active* → Click on *Apply*



8.1.8 Select Step 8 and click on *Edit extension* and use *watonx.ai extension*.

5 is No

7 Thanks for the response .

Action complete

5 is Yes

8 This step has no content

Use an extension

5 is Yes ? is true

9 Jobs splitted : [split_output](#)

Continue to next step

5 is Yes

10 This step has no content

Continue to next step

5 is Yes

New step +

New condition group +

Assistant says

B I @ fx ↻ 📎 🎵 📺 📷 00 </>

For example: What type of transfer would you like to make?

Define customer response

And then

Use an extension

Extension ① Extension not fully configured

Edit extension

8.1.9 Update below fields

Field name	Fetching from
input	Previous step output (Ex: split_task)
Project_id	Project id from watsonx.ai → Step
Model_id	ibm/granite-3-8b-instruct
Version	2023-05-29
Max_tokens	500 & above (based on generated answer)
Min_tokens	0

Continue to next step

5 is Yes

6 This step has no content

Continue to next step

5 is No

7 Thanks for the response .

Action complete

5 is Yes

8 This step has no content

Use an extension

5 is Yes 8 is true

9 Jobs splitted : [split_output](#)

Continue to next step

5 is Yes

10 This step has no content

Continue to next step

5 is Yes

Use an extension

Extension setup

Choose an extension and operation. Then select the information to be shared with the external application to respond to your users' needs. [Learn more](#)

Extension ①

Watsonx Extension

Choose an extension that has been added to your assistant.

Operation ①

Generation

Choose from a list of operations included in your extension.

Privacy ①

☐ Protect data returned from this extension

Parameters ①

Set Tr input To Tr split_task

Set Tr model_id To ibm/granite-3-8b-instruct

Set Tr project_id To 8ac29562-d8a4-4616-b69d-b4f3217fc1da

Set Tr version To 2023-05-29

Optional parameters

Cancel Apply

The screenshot shows a workflow editor with steps 7 through 10. Step 7 is 'Thanks for the response.' with a 'No' condition. Step 8 is 'Splitting the job description' with a 'Yes' condition and a 'Use an extension' button. Step 9 is 'Job splitted into Tasks : [split_output](#)' with a 'Yes' condition and a 'true' condition. Step 10 is 'This step has no content' with a 'Yes' condition. The 'And then' panel shows the 'Use an extension' configuration for 'watsonxai' with the operation 'Generation'. The parameters are set as follows:

Parameter	Value
input	f₁ Expression
model_id	ibm/granite-3-8b-instruct
project_id	8ac29562-d8a4-4616-b69d-b4f321...
parameters.max_new_...	500
parameters.min_new_t...	0
parameters.repetition_...	1
version	2023-05-29

8.1.10 Select Step11 and Click on *Edit extension* and use *watonx.ai extension*. (follow 7.1.8 &7.1.9)

The screenshot shows a workflow editor with steps 10 through 13. Step 10 is 'This step has no content' with a 'Continue to next step' button. Step 11 is 'Processing your request through AI integration' with a 'Yes' condition and a 'Use an extension' button. Step 12 is 'Response : [model_output](#)' with a 'true' condition and a 'Yes' condition. Step 13 is 'Do you want update in Maximo with above steps ?' with a 'Confirmation' button. The 'And then' panel shows the 'Use an extension' configuration for 'watsonxai' with the operation 'Generation'. The parameters are set as follows:

Parameter	Value
input	f₁ Expression
model_id	ibm/granite-3-8b-instruct
project_id	8ac29562-d8a4-4616-b69d-b4f321...
parameters.max_new_...	800
parameters.min_new_t...	0
parameters.repetition_...	1
version	2023-05-29

8.1.11 Select Step 16 and click on *Edit extension* and use Maximo **Maximo Job Plan** extension to update the *status from Active to Pending revised*.

The screenshot shows the 'Extension setup' dialog for the 'Maximo Job Plan Extension'. The operation selected is 'Update job plan status'. The privacy setting is 'Protect data returned from this extension'. The parameters are set as follows:

Set	To
Tr status	PNDREV
Tr pluscrevcom	updating status to pending
pluscrevnum	2 body.pluscrevnum
jobplanid	2 body.jobplanid

And then

Use an extension

Extension ⓘ

Maximo Job Plan Extension

Operation ⓘ

Update job plan status

Parameters ⓘ

status

set to

PNDREV

pluscrevcom

set to

updating status to pending

pluscrevnum

set to

2 7. body.pluscrevnum

jobplanid

set to

2 5. body.jobplanid

Edit extension

8.1.12 select Step 18 and click on *Edit extension* and use Maximo **Maximo Job Plan** extension to **update the job tasks in Maximo**.

Use an extension

Extension setup

Choose an extension and operation. Then select the information to be shared with the external application to respond to your users' needs. [Learn more](#)

Extension ⓘ

Maximo Job Plan Extension

Choose an extension that has been added to your assistant.

Operation ⓘ

Update job plan status

Choose from a list of operations included in your extension.

Privacy ⓘ

☐ Protect data returned from this extension

Parameters ⓘ

Set

Tr status

To

PNDREV

Set

Tr pluscrevcom

To

updating status to pending

Set

pluscrevnum

To

2 7. body.pluscrevnum

Set

jobplanid

To

2 5. body.jobplanid

Cancel

Apply

And then

Use an extension

Extension ⓘ

Maximo Job Plan Extension

Operation ⓘ

update Job Plan

Parameters ⓘ

body

set to

f_x Expression

Edit extension

8.1.13 Select Step 20 and click on *Edit extension* and use Maximo **update_status_maximo** extension to update the *status* from *Pending revised* to *Active*.

Use an extension

✕

Extension setup

Choose an extension and operation. Then select the information to be shared with the external application to respond to your users' needs. [Learn more](#)

Extension ⓘ

Maximo Job Plan Extension

Choose an extension that has been added to your assistant.

Operation ⓘ

Update job plan status

Choose from a list of operations included in your extension.

Privacy ⓘ

☐ Protect data returned from this extension

Parameters ⓘ

Set	Tr status	To	Active
Set	jobplanid	To	2 body.jobplanid
Set	Tr pluscrevcom	To	Updating status to Active
Set	pluscrevnum	To	2 body.pluscrevnum

Cancel

Apply

Extension ⓘ

Maximo Job Plan Extension

Operation ⓘ

Update job plan status

Parameters ⓘ

status	set to	Active
jobplanid	set to	2 5. body.jobplanid
pluscrevcom	set to	Updating status to Active
pluscrevnum	set to	2 7. body.pluscrevnum

Edit extension

8.1.14 Save the *Actions*. → Click on *Close*

IBM watsonx Orchestrate

AI assistant builder

get_jobplan

Editor

Visualization

ⓘ

⚙️

🔍

✕

retrive_job

16 is true

17 Updated status : 16 body.maximo_response.status Jobplanid ...

Continue to next step

18 is true

18 Updating tasks in Maximo

Use an extension

19 is true

19 15 body.message

Continue to next step

Is taken with conditions

Set variable values f.

Conditions 1 condition

If All of this is true:

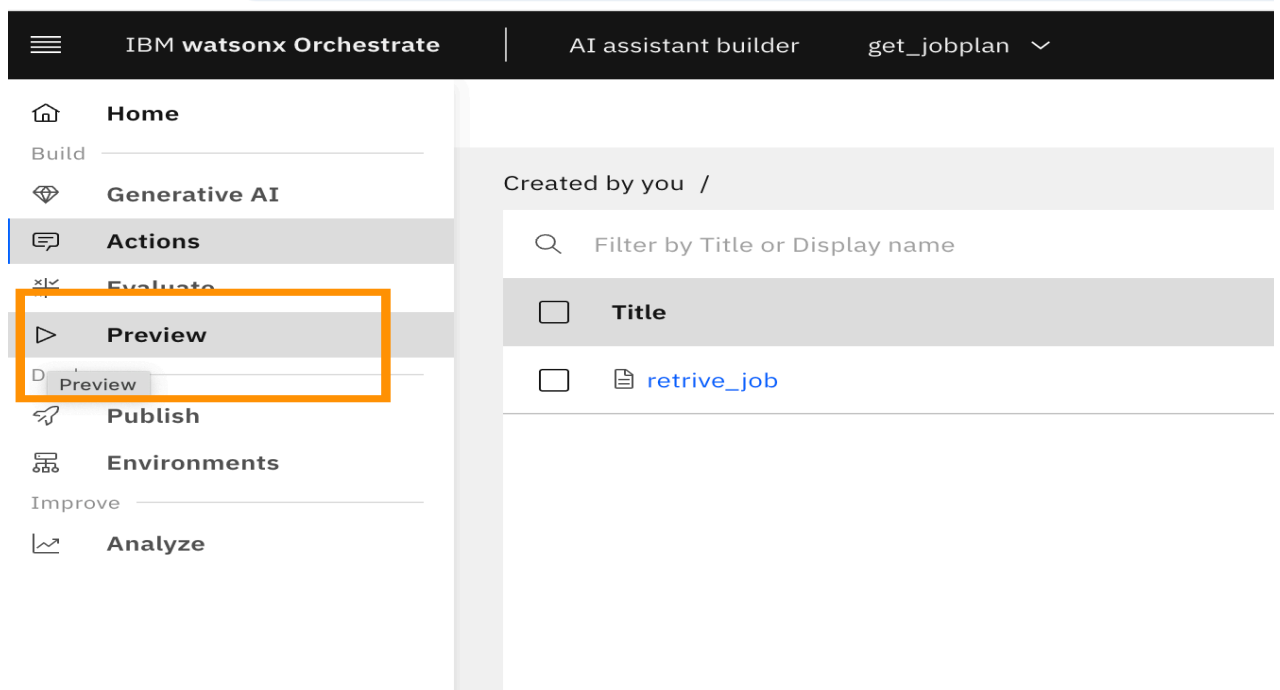
20 Ran successfully == true

and Add condition +

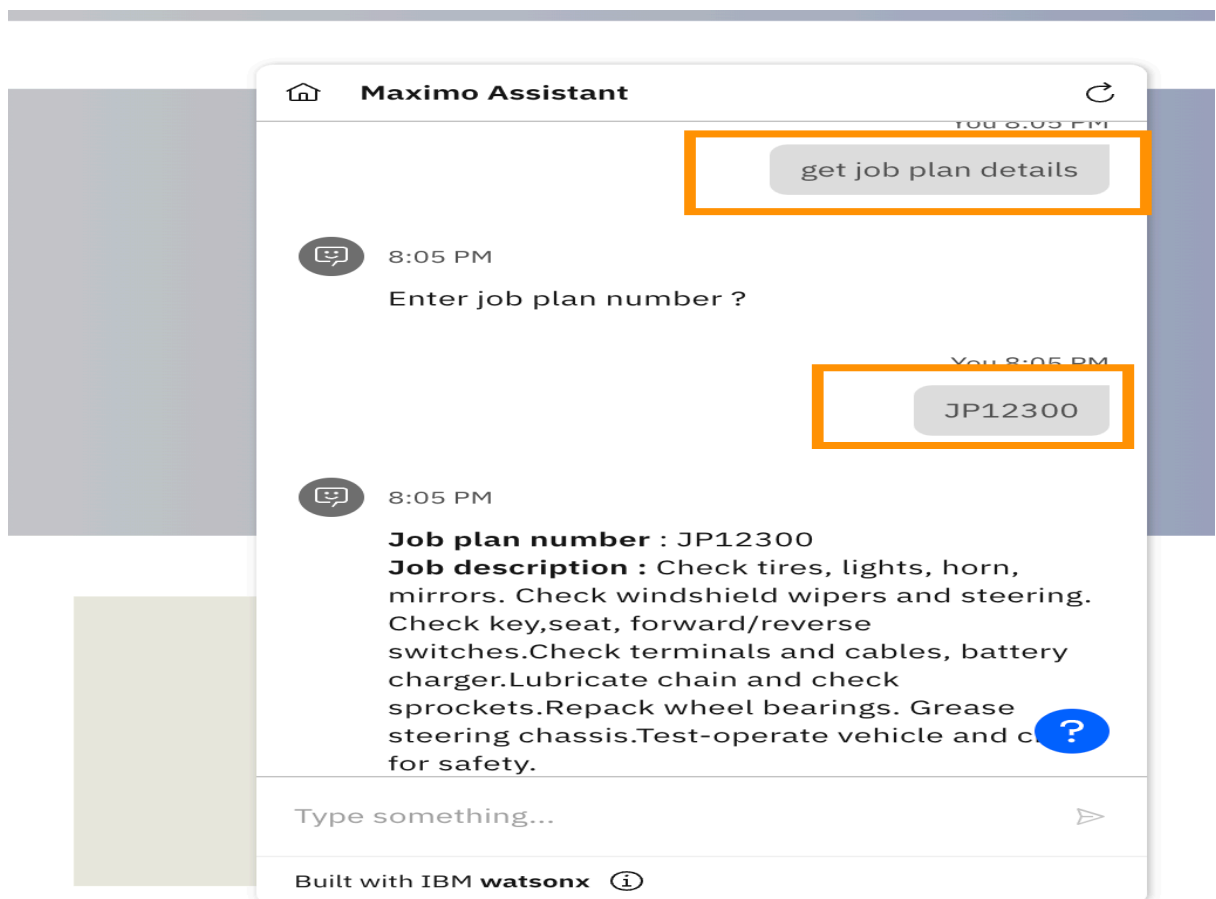
New condition group +

Assistant says

8.1.15 From *left* Navigation Click on *Preview*



8.1.16 Ask Assistant to fetch job plan details and provide job plan number



9. Conclusion

Transforming long, unstructured job plan descriptions into clearly defined job tasks significantly improves the usability and effectiveness of maintenance planning in IBM Maximo.

By automating this process based on the provided job plan number, organizations can:

- Convert complex instructions into simple, actionable steps
- Enhance clarity for technicians executing the work
- Ensure consistency in task execution across teams
- Reduce interpretation errors and operational delays

This approach not only improves field-level execution but also strengthens the integrity and maintainability of job plans in the system. As a result, maintenance teams can operate with greater efficiency, confidence, and alignment to best practices—making Maximo job plans a more powerful tool in day-to-day asset management operations.